Raman Khurana

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Summary

As a seasoned particle physicist-cum-machine learning researcher with extensive experience in both theoretical and applied machine learning, I have spearheaded the publication of four influential papers on dark matter searches involving the Higgs boson at CMS, where I served as the main analyzer and collaborated effectively with fellow scientists. I have independently developed frameworks for several analyses from scratch, creating shareable and scalable code that has been utilized across multiple publications. My technical expertise encompasses the development and implementation of advanced machine learning classifiers, notably enhancing Higgs boson tagging efficiencies. Furthermore, I have significantly improved transformer and MLP-based models for diverse data sources and crafted sophisticated feature engineering framework for streaming data. My contributions also include integrating these models into real-time data processing pipelines using industry-standard big data tools such as Apache Kafka and Flink, optimizing for effective inference. I am eager to leverage my skills to advance the collaborative initiatives of UW-Madison's physics and Data Science Institute, contributing to groundbreaking research and development.

Experience

Staff Researcher, Northwestern University, USA *September 2022 - Present*

- Enhanced transformer and MLP-based models for robust time series forecasting by using static dataset source (Paper in progress). Extformer
  - Experimented different strategies to integrate the architecture to process static data (in PyTorch).
  - Designed a K8 cluster and integrated a GitHub CI/CD pipeline for Docker image deployment.
- Supervising a PhD for development an automated feature engineering model for streaming data.
  - Crafted a strategy for processing streaming data and actively contributed to technical discussions.
- Supervised team of Master DS students for practicum for development of ML models to predict wild fire using edge devices. CDL-Wildfire
- Development and maintenance of real time data processing pipeline for ML model inference using industrial tools like, Apache Kafka, Flink, Grafana. REFIT

Research Associate, National Taiwan University *April 2022 - September 2022*

- Led the development of a data-driven background estimate and development of framework for upper limit estimation for BSM Higgs boson using same sign top pair production. **PLB 850 (2024) 138478**

Postdoctoral Researcher, Florida State University, USA *September 2018 - March 2022*

- Initiated and spearheaded a new search for dark matter at CMS using non-resonant production of bottom quarks. Presented this search at workshop and important occasions. **CMS-PAS-SUS-23-008**
  - Conceived and successfully implemented the project from initial concept to successful execution.
  - Early adopter of pandas and columnar analysis techniques to cut computation time (pre-uproot).
  - Comparative and early development using awkward and iris-hep tools. AwkwardAnalyzer.
- Leadership in Mono-Higgs combination at CMS. [Analysis contact] **JHEP 03 (2020) 025**
  - Envisioned and led the statistical combination of five Higgs decay modes, enhancing research scope.
  - Organized efforts to ensure effective collaboration across teams, preventing data overlap and standardizing systematic treatments

- Presented this important combination for pre-approval, approval and conferences.

- CMS Electromagnetic Calorimeter operations and advanced Analysis of ECAL L1 Trigger Pre-firing.
  - Quantified the relative impacts sources causing ECAL L1 trigger pre-firing during Run 2 using simulation and special data collected to perform measurement.
  - Extending the project to emulate effects in simulations and enhance control and monitoring (in DQM) for Run 3.

Postdoctoral Researcher, National Central University, Taiwan *March 2015 - September 2018*

- Extend and re-stratagise the search for dark matter produced in associated with Higgs boson using $36 fb^{-1}$ of data. [Analysis contact] **Eur Phys J C 79 (2019) 280**.
  - Development of data-driven background estimation using control regions for major backgrounds.
  - Showcased enhanced H-tagging performance with CA15 jets and energy correlation variables.
  - Inclusion of ML approach to discriminate boosted Higgs boson ($b\bar{b}$; double b-tagger) from QCD.
- Leadership in the first dark matter search using Higgs boson as a tool. **JHEP 10 (2017) 180**.
  - Explored mono-H (bb) signatures using a 2.3 fb^{-1} dataset and restricted the allowed phase-space.
  - Engineered robust analysis framework using C++, integrating findings with mono-H ($\gamma\gamma$).
  - Presented analysis for pre-approval, approval, and international conferences. [Analysis contact]
- Controlling the anomalous p_T^{miss} events in simulation and early Run 2 data. **JINST 14 (2019) P07004**
  - Studied the effect of detector inactive regions on anomalous p_T^{miss} events using simulation.
  - Investigated the impact of reconstruction and instrumental factors on p_T^{miss} tails and developed an efficient tagging filter.

PhD, Saha Institute of Nuclear Physics, India *August 2009 - March 2015*

- Lead the search for Higgs boson in $H \to ZZ \to 2l2\tau$ final state using CMS and contributed towards combination of all $H \to ZZ \to 4l$ searches along with measurement of production cross-section for $ZZ \to 4l$ process. [Analysis Contact] **PRD 89 (2014) 092007**, **PLB 721 (2013) 190**, **JHEP 03 (2012) 081**
  - Crafted a C++ analysis framework for feasibility studies using simulation and initial collision data.
  - Worked on the data-driven background estimation method and validated the measurement using simulation.
  - Presented analysis for pre-approval, approval and International conferences.
- Ehhancing hadronic tau (τ_h) reconstruction and identification efficiency. **JINST 11 (2016) P01019**
  - Optimization of τ_h-id for high pileup, high-p_T reconstruction resulting in 5-50% improvement.
  - Measurement of $e \to \tau_h$ mis-identification using tag and probe.

Other Contributions

- **Geant4 Simulation and Analysis for High Granular Calorimeter (HGCAL) test beam data.**
  - Established the first standalone simulation setup for HGCAL, conducting key studies on active materials, resolution, and Moliere radius.
  - Utilized test beam data from HGCAL to demonstrate time resolution in Si sensors to distinguish between hard and pileup interactions. **JINST 13 (2018) 10, P10023**.
- **Leadership of Jet and p_T^{miss} Data Quality Monitoring Group to safeguard data quality.**
  - Led a team of 12 physicists for quality checks on CMS's recorded data collection of $\approx$100 fb^{-1}.
  - Validated software updates with simulated and real data to ensure system reliability.
  - Developed a new tool for real-time monitoring of Jet and p_T^{miss} variables.

Mentoring Experience

Ms. Holly Wang at Northwestern University, USA: Project: Development of automated feautre engineering tool for streaming data (in progress).

CMS DAS 2021 at Fermilab, USA: Facilitator/tutor for hadronic tau τ_h reconstruction and Z$\rightarrow$ $\tau\tau$ cross-section measurement in 2 week virtual CMS Data Analysis School (CMS DAS) organised by Fermilab.

Mr. Deepak Kumar, PhD, IISc, Banglore. Thesis project: Search for dark matter produced in association with Higgs boson decaying into bottom quarks using full Run 2 data. Status: Completed.

Mr. Praveen Chandra Tiwari, PhD, IISc, Banglore. Thesis project: Search for dark matter produced in association with non-resonant production of pair of bottom quarks. Status: Completed.

Ms. Pampa Ghose, PhD, FSU, USA. Project: Study of ECAL L1 Trigger pre-firing using simulation data. Status: Completed.

Mr. Kung-Hsiang, master student, National Central University. Thesis: Study of substructure variable for mono-Higgs$\rightarrow$bb analysis. Status: Completed.

Ms. Fasya Khuzaimah, master student, National Central University. Thesis: Study of double b-tagger variable and derivation of data to simulation scale factors for mono-Higgs$\rightarrow$bb analysis.

Mr. Ching-Wei Chang, master student, National Central University. Thesis: Search for new resonance decaying into pair of Higgs boson (HH$\rightarrow$ 4b). Status: completed.

Ms. Fang-Ying Tsai, master student, National Central University. Thesis: Search for dark matter produced in association with Higgs boson decaying into bottom quarks (mono-Higgs$\rightarrow$bb). Status: completed.

Research Honors

Ramanujan Fellowship 2021, with a $150K research grant to conduct the search for dark matter using Higgs boson.

Senior Research Fellowship, CSIR, India 2011.

Junior Research Fellowship, CSIR, India 2009.

Education

PhD in Experimental High Energy Physics
University of Calcutta and Saha Institute of Nuclear Physics, India *2014*

MSc in Physics (Electronics Specialization),
University of Delhi, India *2009*

Presentations at International conferences

Invited/Plenary/Parallel Presentation

- Search for Dark Matter using Higgs boson as a tool. ICHEP 2022.
- Search for dark matter at CMS, Virtual World, ICHEP 2020.

- Searches for dark matter and long-lived particles at the LHC, Cancelled due to Covid 19, La Thuile, Italy, 2020.
- Dark Matter searches at LHC, LHCP2017, Shanghai, China.
- Dark Matter searches at LHC, Warsaw Workshop on Non-Standard Dark Matter 2016, Poland.
- Search for heavy resonances in the W/Z-tagged dijet mass spectrum at CMS, PSROC 2016, Kaohsiung city, Taiwan.
- Search for standard model Higgs boson in tau final states, Workshop on Higgs and BSM physics 2013, ICTP, Italy.
- Search for Higgs boson in $H \to ZZ \to 2l2\tau$ decay channel presented at XX DAE-BRNS High Energy Physics Symposium 2013, Visva-Bharati, India.

Posters

- Investigation of the fast timing capabilities of the Silicon sensors for the CMS high granularity calorimeter, LHCP 2017, Shanghai, China.
- Identification of hadronic tau decays in CMS, PANIC 2014, Hamburg, Germany. **JINST 11 (2016) P01019**.
- Search for Standard Model Higgs Boson in $H \to ZZ \to 2l2\tau$ decay channel with CMS detector at $\sqrt{s}$ = 7 and 8 TeV, EPS HEP 2013, Stockholm.
- Tau reconstruction and identification at CMS, ICHEP 2012, Australia. **PoS ICHEP2012 (2013) 518**
- Tau identification commissioning results, Lepton Photon 2011, Mumbai, India. **Pramana 79 (2012) 1235-1238**

Workshop Participation

- ECAL L1 Trigger pre-firing studies. ECAL days in Zurich 2019, Zurich, Switzerland.
- Search for dark matter produced in association with Higgs boson using Run 2 data., CMS Exotica Workshop, Greece.
- Search for dark matter in Higgs ($H \to b\bar{b}$) + p_T^{miss} final state, CMS Exotica workshop 2015, Venice, Italy.

Publications

Full list of publication can be accessed at Inspire-hep. I list here my key publications where I was leading the analysis/study or I have contributed significantly to the publication.

1. Search for dark matter produced in association with a pair of bottom quarks in proton-proton collisions at $\sqrt{s}$=13 TeV, **CMS-PAS-SUS-23-008**
2. Search for new Higgs bosons through same-sign top quark pair production in association with a jet in proton-proton collisions at $\sqrt{s}$ = 13 TeV. **Phys. Lett. B 850 (2024) 138478**
3. Search for dark matter particles produced in association with a Higgs boson in proton-proton collisions at $\sqrt{s}$ = 13 TeV. **JHEP 03 (2020) 025**.
4. Performance of missing transverse momentum reconstruction in proton-proton collisions at $\sqrt{s}$ = 13 TeV using the CMS detector **JINST 14 (2019) P07004**.
5. Search for dark matter produced in association with a Higgs boson decaying to a pair of bottom quarks in proton–proton collisions at $\sqrt{s}$=13 TeV. **Eur. Phys. J. C 79 (2019) 280**.

6. Search for a massive resonance decaying to a pair of Higgs bosons in the four b quark final state in proton-proton collisions at $\sqrt{s} = 13$ TeV,. **Phys. Lett. B 781 (2018) 244**.
7. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos and b quarks at $\sqrt{s} = 13$ TeV. **JHEP 11 (2018) 172**.
8. First beam tests of prototype silicon modules for the CMS High Granularity Endcap Calorimeter **JINST 13 (2018) 10, P10023**.
9. Search for associated production of dark matter with a Higgs boson decaying to $b\bar{b}$ or $\gamma\gamma$ at $\sqrt{s} = 13$ TeV. **JHEP 10 (2017) 180**.
10. Search for heavy resonances decaying into a vector boson and a Higgs boson in the ($\nu\nu$, ν, ll) $b\bar{b}$ final states. **Phys. Lett. B 768 (2017) 137**.
11. Reconstruction and identification of τ lepton decays to hadrons and ν_τ at CMS. **JINST 11 (2016) P01019** .
12. Search for a Higgs boson in the mass range from 145 to 1000 GeV decaying to a pair of W or Z bosons. **JHEP 10 (2015) 144**.
13. Properties of the Higgs-like boson in the decay $H \to ZZ \to 4l$ in pp collisions at $\sqrt{s} = 7$ TeV and 8 TeV. **Phys. Rev. D 89 (2014) 092007**.
14. Evidence for the direct decay of the 125 GeV Higgs boson to fermions **Nature Physics 10 (2014) 557**.
15. Measurement of the W^+W^- and ZZ production cross sections in pp collisions at $\sqrt{s} = 8$ TeV. **Phys. Lett. B 721 (2013) 190**.
16. Measurement of the ZZ production cross section and search for anomalous couplings at $\sqrt{s} = 7$ TeV. **JHEP 01 (2013) 063**.
17. Search for a standard-model-like Higgs boson with a mass in the range 145 to 1000 GeV at the LHC. **Eur. Phys. J. C 73 (2013) 2469**.
18. Updated results on the new boson discovered in the search for the standard model Higgs boson in the $H \to ZZ \to 4l$ channel in pp collisions at $\sqrt{s}$ =7 and 8 TeV. **Phys. Rev. Lett. 110 (2013) 081803**.
19. Search for a Standard Model Higgs boson produced in the decay channel $H \to ZZ \to 2l2\tau$ with CMS detector at $\sqrt{s}$=7 TeV,. **JHEP 03 (2012) 081**.
20. Dark Matter Benchmark Models for Early LHC Run-2 Searches: Report of the ATLAS/CMS Dark Matter Forum. **Phys. Dark Univ. 26 (2019) 100371**.

Seminar

National Central University, Taiwan *March 2014*
Search for Higgs boson in H→ZZ→2l2τ final state using CMS detector at CERN.

Kansas State University, USA *April 2016*
Search for dark matter produced in association with Higgs boson decaying to a pair of bottom quarks using CMS data collected in 2015.

Tata Institute of Fundamental Research, India *June 2022*
Search for dark matter produced in association with resonant (Higgs boson) or non-resonant production of pair of bottom quarks.

Northwestern University, USA (AI Journal Club Seminar): *May 2023*
Comparative analysis of statistical, machine learning and deep learning models for time series forecasting using open source datasets.

Technical Skills

- **Programming Languages:** Python, C++, Shell Scripting
- **Data Analysis Tools:** ROOT, GEANT4, PYTHIA, MADGRAPH
- **Machine Learning Frameworks:** Scikit-Learn, TensorFlow, Keras, PyTorch
- **Statistical Tools:** RooFit, RooStats, combine
- **Visualization Software:** ROOT, Matplotlib, Seaborn
- **Other Technologies:** Docker, Kubernetes, GitHub CI/CD, FPGA, Raspberry Pi